

SYM, Russian Federation

WMO: 239750

Lat: 60.3472N

Lon: 88.3636E

Elev: 281

StdP: 14.55

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -53.3	(c) -46.2	(d) -56.9	(e) 0.2	(f) -53.2	(g) -50.6	(h) 0.3	(i) -46.1	(j) 18.1	(k) 14.3	(l) 15.7	(m) 7.3	(n) 0.4	(o) 310	(p) 0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.1	(c) 85.3	(d) 65.4	(e) 81.8	(f) 64.3	(g) 78.0	(h) 62.8	(i) 68.8	(j) 79.2	(k) 66.6	(l) 76.7	(m) 64.8	(n) 74.7	(o) 5.1	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 65.4	(b) 94.6	(c) 72.6	(d) 63.2	(e) 87.5	(f) 70.7	(g) 61.0	(h) 80.9	(i) 68.7	(j) 33.3	(k) 79.2	(l) 31.5	(m) 76.8	(n) 30.0	(o) 74.6	(p) 80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	13.7	11.7	-55.8	91.6	9.2	3.3	-62.4	94.0	-67.8	95.9	-72.9	97.7	-79.6	100.1
			-56.2	72.5	9.0	3.4	-62.7	75.0	-67.9	76.9	-73.0	78.8	-79.6	81.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	-11.6	-2.2	14.4	29.2	43.9	59.8	64.7	57.5	45.2	29.5	5.4	-6.5
	DBStd	29.20	19.23	15.89	13.56	10.65	9.74	8.92	6.40	6.65	7.86	11.21	18.61	19.34
	HDD50	9291	1909	1460	1104	624	247	22	0	12	187	637	1339	1750
	HDD65	13786	2374	1880	1569	1074	657	200	85	246	596	1101	1789	2215
	CDD50	1119	0	0	0	0	57	315	458	246	42	1	0	0
	CDD65	137	0	0	0	0	2	43	77	14	1	0	0	0
	CDH74	1755	0	0	0	0	35	649	885	180	6	0	0	0
	CDH80	496	0	0	0	0	5	194	257	40	0	0	0	0
Wind	WSAvg	4.3	3.6	4.2	4.9	5.3	5.2	4.1	3.5	3.6	3.9	4.7	4.3	4.1
Precipitation	PrecAvg	20.9	1.1	0.9	0.9	1.0	1.7	2.5	2.7	2.6	2.6	2.0	1.7	1.5
	PrecMax	31.7	2.4	1.9	1.6	2.3	5.6	7.6	5.1	6.3	4.2	3.6	3.1	4.2
	PrecMin	13.3	0.1	0.4	0.2	0.2	0.7	0.4	0.1	0.5	0.8	0.6	1.1	0.6
	PrecStd	4.1	0.6	0.4	0.4	0.5	1.0	1.3	1.3	1.4	0.9	0.7	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	31.3	46.8	60.8	78.8	90.2	90.7	85.2	74.5	56.0	38.0	31.1
		MCWB	29.1	30.0	37.9	47.2	60.2	66.1	67.7	66.4	59.2	49.0	35.7	29.6
	2%	DB	24.6	27.7	40.4	54.6	71.9	85.3	85.9	78.8	68.0	51.4	34.8	27.5
		MCWB	23.5	26.2	33.3	43.8	55.9	64.2	66.3	64.1	56.6	46.0	33.7	26.3
	5%	DB	19.9	23.9	36.4	49.5	66.9	81.5	82.5	74.6	62.6	47.1	33.0	24.3
		MCWB	18.8	22.5	31.8	41.2	53.8	63.0	65.3	62.1	54.4	43.4	31.9	23.2
	10%	DB	14.6	19.3	33.1	45.0	61.3	77.3	78.9	70.5	58.0	43.0	30.4	20.0
		MCWB	13.7	17.7	29.4	38.0	50.2	61.4	63.9	60.2	52.2	40.0	29.2	18.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.1	30.4	38.6	48.9	61.8	70.3	72.5	69.6	61.5	50.4	35.9	29.9
		MCDB	29.7	31.2	46.1	56.8	76.6	84.1	83.4	79.4	70.5	54.0	36.7	30.9
	2%	WB	23.6	26.4	34.7	45.0	57.5	66.8	69.8	66.0	58.0	46.7	34.0	26.4
		MCDB	24.5	27.5	38.8	53.1	68.8	78.9	80.1	76.5	65.9	50.6	34.7	27.3
	5%	WB	18.9	22.5	32.2	41.4	54.7	64.9	67.8	63.5	55.4	43.6	31.8	23.3
		MCDB	19.9	23.9	36.4	48.0	64.9	76.8	77.6	71.6	61.1	46.7	32.9	24.3
	10%	WB	13.6	17.8	29.7	38.6	51.1	63.0	66.1	61.3	52.9	40.5	29.2	18.9
		MCDB	14.5	19.2	33.2	44.4	60.8	74.1	75.2	68.7	57.6	42.9	30.4	19.9
Mean Daily Temperature Range	5% DB	MDBR	18.2	20.8	23.0	19.2	18.3	20.4	20.1	17.4	14.3	11.4	15.3	16.7
		MCDBR	15.7	17.5	23.1	25.4	27.1	30.7	27.8	26.2	23.6	15.5	9.5	13.5
		MCWBR	15.6	16.4	18.2	17.1	15.0	13.1	11.9	13.5	13.9	11.1	9.1	13.2
	5% WB	MCDBR	15.9	17.2	21.6	22.1	24.7	24.4	22.3	21.5	20.3	14.1	9.8	13.5
		MCWBR	15.7	16.6	17.7	16.1	14.6	12.1	11.8	12.3	12.7	10.5	9.8	13.2
Clear-Sky Solar Irradiance	taub		0.271	0.306	0.343	0.368	0.387	0.399	0.467	0.413	0.360	0.327	0.272	0.252
	taud		1.946	1.986	1.978	2.056	2.264	2.340	2.111	2.261	2.375	2.254	2.004	1.893
	Ebn at Noon		137	205	239	259	265	264	239	244	239	202	136	92
	Edh at Noon		16	27	38	43	38	36	44	35	26	21	14	11
All-Sky Solar Radiation	RadAvg		139	429	886	1377	1555	1762	1636	1187	706	318	151	71
	RadStd		17	32	76	140	170	196	141	108	101	43	32	11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page